ZOO502 MID TERM

Subjective

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Regards VUWAYS Team

Welcome To vuways Study Help

1. **Define kinesiology (2)**

It is the Study of muscles, lever systems, and their working

2. **Name the parts of membranous labyrinth?(2)**

Membranous Labyrinth: It consists of three parts:

1) Three semicircular canals 2) Vestibule (sacculus and utricles) 3) Cochlea

3. **What is ryanodine receptors?(2)**

The membrane of SR in the triad region has specialized Ca2+ ion channel proteins called ryanodine receptors.

4. **What are triads(2)**

In the skeletal muscle fiber, a T tubule is associated with two terminal cisternae of sarcoplasmic reticulum on its both sides this arrangement of three associated tubes or sacks forms a structure that is called a triad

5. **Define term cardiac output and stroke volume(2)**

Cardiac Output • "Quantity of blood pumped into the aorta each minute by the heart". Average cardiac output for resting adult human: 5 L/min. **Stroke Volume:** "Volume of the blood ejected by each beat of heart

6. **Kemography 4 uses? (2)**
    1. Study of skeletal muscle contractions (twitch and tetanus)
    2. Cardiac muscle activities (cardiac cycle)
    3. Measurement of blood pressure
    4. Rate of respiration
7. **Components of Rhodopsin(2)**

A rhodopsin molecule consists of 2 major components: • Opsinprotein • A light-absorbing molecule (retinal or 3dehydroretinal)

8. **Pacemaker what type of cells induce?(2)** Ans:

9. **What is sarcolemma and sarcoplasma? (2)**

Ans: The membrane of the cell is called sarcolemma. Its cytoplasm is called sarcoplasm.

10. **Myofilament and its types (2)**

Ans: **Myofilaments**

- A myofibril is composed of myofilaments. • Myofilaments are of two types:
- Thin filaments composed of actin
- Thick filaments composed of myosin

3 marks

11. **Define Pacemaker and its types**

Pacemakers • The excitatory region. Generates rhythmical impulses. • Control the rhythmicity of cardiac chambers. **Pacemaker Types**: Two basic types of pacemakers in animals with pumping hearts:

• Neurogenic pacemakers • Myogenic Pacemakers

Neurogenic Pacemakers • Consists of neurons. • Found in many invertebrate hearts. • Such hearts are neurogenic hearts.

Myogenic Pacemakers • Consists of specialized self-excitatory muscle cells. • Found in some invertebrate & all vertebrate hearts. • Such hearts are known as myogenic hearts

12. **Myofilamentand its types 3**

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A myofibril is composed of myofilaments.**Two types**: • Thin filaments composed of actin • Thick filaments composed of myosin

13. **Name 3 nonhuman sensory modalities(3)**

Many animals possess other sensory modalities that are unavailable to human beings.

Examples • Pit organs • Electroreceptors • Magnoreceptors

14. **Role of T-Tubules 3**

T tubule system provides the anatomic link between the surface membrane and the myofibrils deep inside the muscle fiber. • When an action potential spreads over a muscle fiber membrane, a potential change also spreads along the T tubules to the deep interior of the muscle fiber. • In the cell interior, T tubes are linked to SR. • SR releases Ca2+ that permit myosin cross-bridges to attach and generate force for contraction

15. **General Features of smooth Muscles 3** Least specialized muscle fibers.
    - Have myosin similar to that found in contractile nonmuscle cells.
    - Non-striated. • Involuntary: under autonomic control.
    - Contract and relax slowly.
    - Capable of more sustained contractions
16. **function of transduction 3**

The physical or chemical energy of stimulus is converted into electrical signals i.e nerve impulse by sensory receptor cells. This conversion is called sensory transduction.

17. **write the signaling system of cAmp**

Signaling System of cAMP: Many hormones use the AdenylateCyclase–cAMP second messenger signaling system to stimulate their target tissues. • The hormone binds to the receptor which is coupled to a G protein. G protein stimulates adenylatecyclase in the membrane. Activated AC catalyzes the conversion of ATP into cAMP in cytoplasm.

• cAMP then activates an enzyme called cAMPdependent protein kinase. • Protein kinase phosphorylates specific proteins in the cell, triggering a cascade of biochemical reactions that ultimately lead to cell's response to the hormone.

18. **How many group the second messenger, lipid insoluble hormone is devided?3**
    - Cyclic nucleotide monophosphates e.g. cAMP and cGMP.

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- Inositol phospholipids e.g. inositol trisphosphate (InsP3) and diacylglycerol (DAG)
- Ca2+ ions and associated calmodulin

19. Different parts of membranous labyrinth?(3)

Ans: Membranous Labyrinth: It consists of three parts:

- Three semicircular canals • Vestibule (sacculus and utricles) • Cochlea
- Cochlea is the sensory structure concerned with hearing.
- It transduces the mechanical energy of sound vibrations into electrical nerve impulse.

20. Differentiate tetanus and muscles twitches?(3)

Ans: **Tetanus** • A sustained muscle contraction evoked by stimulation from simultaneous multiple impulses is known as tetanic contraction. • During tetanized state, the contracting tension in the muscle remains constant in a steady state

21. Effects of transduction (3)

Ans:

22. Effects of LH on male and female (3)

Ans: Function of LH in male and female. In Females LH stimulates production of estrogen and progesterone by ovary. It causes ovulation and formation of corpus luteum. In males, LH is also called as ICSH (interstitial cell stimulating hormone) It stimulates testosterone production by the Leydig cells of testes.

23. Gonadotropin (3)

Ans: LH and FSH are referred to as gonadotropins.

- They act on the gonads and stimulate secretion of gonadal hormones. It also stimulates secretion of androgens by adrenal cortex.
- It plays role to maintain the size of endocrine tissues (zona fasciculata & zona reticularis) of adrenal cortex.
- It also stimulates lipoprotein uptake into cortical cells for the biosynthesis of cholesterol and steroid hormones.
- In many organisms, ACTH also plays role in circadian rhythms.

24. feed back mechanism and its principle and significance(3)

Ans: A feedback mechanism connects the response to the initial stimulus through a feedback loop or circuit. • Feedback loops are characteristic of controlled pathways of hormones. • Through these loops, secretion of hormone is regulated.

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Principle: Output or product itself regulates the process
Significance: Maintenance of homeostasis
Mechanism: Feedback controls regulate particular variables e.g., • Temperature • Salinity • pH This regulation requires:

- Continuous sampling of controlled variables
- Respective corrective actions.

25. Meissner's corpuscles and pacinian corpuscles difference? (3)

Meissner's Corpuscles: are the touch receptors. They have encapsulated nerve endings which lie in papillae which extend into the ridges of fingertips.

Pacinian corpuscles: are situated quite deep in the skin. They have encapsulated nerve endings and receive deep pressure stimulus

5 marks

26. Write a note on structure of scroplasmic technique?(5)

Ans: **Sarcoplasmic Reticulum**

- SR is a network of membrane-bound tubules extending throughout muscle cells on either side of a Z disk and extends from one Z disk to the next as well.
- In many features, it is similar to the ER of other cells.
- The SR has a special organization that is extremely important in controlling muscle contraction.
- Rapidly contracting types of muscle fibers have extensive network of sarcoplasmic reticulum.

Structure of SR: Composed of two major parts:
1- Terminal cisternae and, 2- Longitudinal tubules

- Terminal cisternae are larger chambers that are closely associated with T-tubules (12 nm apart).
- These are the primary site of calcium release as they have channel proteins that open with AP in T tubules.
- Long longitudinal tubules run between the terminal cisternae and surround the myofibrils.
- These are the locations where ion channels for Ca+ ion absorption (Ca+ pumps) are most abundant.
- So, they are involved in Ca2+ sequestering activity.

27. Write difference btw Stigma and ocelli (5)

Eyespot or Stigma • Simplest photoreceptive structure. • Found in some protozoa e.g. euglena. • It is a bright red colored organelle and has carotenoid pigments. • It gives a sense of light and dark. • Helps in phototaxis

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Eyecups or Ocelli • Multicellular photoreceptive structures that consist of a cuplike depression containing photoreceptor cells. • Found in cnidarians and flatworms e.g. Planaria. • Cannot form image. • Provide the animal a sense of direction only.

28. Exocrine hormones with example (5)

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body.

• The fluid secretions may be proteins (enzymes) or mucous or both.

Examples of Exocrine Glands

• Salivary glands produce saliva that is delivered to the oral cavity through submandibular and parotid ducts.

• Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct •Lacrimal glands produce tears that are delivered through lacrimal duct on the surface of eye.

• Mammary glands produce milk that is delivered through lactiferous ducts to the nipples

29. Define Taste buds and explain taste receptor cells (5)

The gustatory organs of vertebrates are called taste buds. Gustatory (Taste) Receptors • Respond to dissolved molecules that come in direct contact with the receptive structure

Taste buds • A taste bud is composed of about 50 types of modified epithelial cells, including: • Supporting cells (sustentacular cells) • Basal cells • Taste receptor cells • basal cells are progenitor cells that give rise to new taste receptors. • They regularly generate new sensory taste receptor cells which have an active life of only 10 days

Taste Receptor Cells • Outer tips of taste cells are arranged around a minute taste pore. • At the tip of taste cell, several microvilli (taste hairs) protrude into the taste pore. • Microvilli approach the cavity of the mouth. • They provide receptor surface for taste. • A branching terminal network of taste nerve fibers surrounds the base of taste cells

30. Conductive system of Heart. 5

- A specialized system for generating rhythmic electrical impulses.
- A system to conduct these impulses rapidly throughout its muscles.
- These impulses cause and control rhythmic contraction of cardiac chambers.

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Effects of Rhythmic Electrical Impulses Rhythmicity of impulses cause:

- Contraction of atria about 1/6th sec. ahead of ventricular contraction.
- Filling of ventricles before they contract.
- Contraction of all portions of ventricles simultaneously

Components of Conductive System: Include:

1. Atrial internodal pathways (conduct impulse from S-A node to the A-V node).
2. A-V node (conducts impulses into the ventricles)
3. A-V bundle (bundle of His), conducts impulses from atria into the ventricles.
4. Left & right branches of A-V bundle.
5. Purkinje fibers, branch off from bundle branches (conduct impulse to all parts of the ventricles).

31. Explain the structure of sarcoplasmic reticulum?5

Structure of SR: Composed of two major parts: 1- Terminal cisternae and, 2- Longitudinal tubules • Terminal cisternae are larger chambers that are closely associated with T-tubules (12 nm apart). • These are the primary site of calcium release as they have channel proteins that open with AP in T tubules.

• Long longitudinal tubules run between the terminal cisternae and surround the myofibrils. • These are the locations where ion channels forCa+ ion absorption are most abundant. • So, they are involved in Ca2+ sequestering activity

32. Negative and positive feedback mechanism 5

Negative Feedback: Secretory activities of most endocrine tissues are modulated by negative feedback, especially those involved in maintaining homeostasis. • In this type of feedback, the hormone itself or its products tend to suppress its further release. • This mechanism ensure a proper level of hormone activity at the target tissue and prevents its overactivity by over-secretion.

Loops oNegative Feedback • -ve feedback involves either a short loop or long loop. • In short-loop, hormone or a byproduct of its activity, acts back directly on the secreting endocrine tissue. • Long-loop feedback includes more than one endocrine gland and hormone e.
In +ve feedback, the secretion of a hormone directly or indirectly leads to its increased secretion. This happens when an extremely rapid or strong response is required. +ve feedback is also common in the early phases of response in most cases of hormonal action.

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33. Write tetanus and it's types

Tetanus • A sustained muscle contraction evoked by stimulation from simultaneous multiple impulses is known as tetanic contraction. • During tetanized state, the contracting tension in the muscle remains constant in a steady state. **Types of Tetanus**

1. A tetanic contraction can be either: • Unfused (incomplete) • Fused (complete).

UnfusedTetanus • An unfused tetanus occurs when the muscles are being stimulated at a faster rate. The fibers do not completely relax before the next stimulus

Fused Tetanus • When there is no relaxation of the muscle fibers between stimuli and the twitches overlap. It occurs during a high frequency of stimulation. A fused tetanic contraction is the maximal possible contraction

34. Interoceptive receptors 5

Interoceptive Receptors These include:

- **Receptors of vestibular system** that monitor orientation of the body. Present in semicircular canals in inner ear.
- **Thermoreceptors** that keep track of thermal state of the body. They are found scattered on the skin.
- **Chemoreceptors** that keep track of chemical state of the body. Have various types distributed in specific parts of the body.
- **Proprioceptors** that monitor position of the muscles and joints. They are located in the movable joints
- **Nociceptors** perceive the sensation of pain. They are distributed evenly in the skin and deeper body parts.
- **Pacinian corpuscles** receive pressure stimulus. They are situated deep in the body. In the limbs, they receive vibrations
- Electrical transmission has been discovered in: • Many cells of central nervous system • Retina of the eye • Smooth muscle fibers • Cardiac muscle fibers • Most receptor cells • Some axons of PNS.

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- • Sensory detection of physical stimuli that have mechanical energy. • e.g. stretch, touch, pressure, sound and gravity
- Taste Receptors in Terrestrial Vertebrates • Located at the anterior region of digestive tract i.e.
- On the tongue • Epiglottis • Back of the mouth • Pharynx • Upper esophagus

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